



Problem-Solution Fit

Date	3 July 2026
Team ID	XXXXXXX
Project Name	Rising Waters - Flood Prediction System
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] Disaster Management Authorities, Meteorologists, Government Agencies, Local Administration, People living in flood-prone areas.	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] Limited budget, limited technical knowledge, lack of advanced forecasting tools, unreliable internet connectivity in some areas.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional forecasting: Easy to use but less accurate. Weather monitoring systems: Useful but expensive and may not provide timely flood predictions.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] Floods occur frequently during monsoon seasons. Existing prediction methods	9. PROBLEM ROOT / CAUSE [RC] Heavy rainfall, poor flood forecasting methods, delayed warnings,	7. BEHAVIOR + ITS INTENSITY [BE] Users regularly monitor weather conditions during the rainy season and seek quick,

are often delayed or less accurate, leading to property damage and loss of life.	inadequate disaster preparedness, and climate variations.	reliable flood predictions before taking action.
3. TRIGGERS TO ACT [TR] Heavy rainfall, increased cloud cover, weather alerts, rising river levels, and monsoon forecasts.	10. YOUR SOLUTION [SL] Develop a Flask-based machine learning web application using a Decision Tree model to predict flood risk from weather parameters and provide timely alerts for better disaster preparedness and resource planning.	8. CHANNELS OF BEHAVIOR [CH] ONLINE Online: Web application, government disaster management portals, weather websites. Offline: Local authorities, emergency response teams, community announcements.
4. EMOTIONS BEFORE / AFTER [EM] Before: Fear, uncertainty, anxiety. After: Confidence, preparedness, and improved decision-making.		OFFLINE